

Agentic AI for Optimizing FAISS HNSW Vector Database Parameters

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Abstract

Hierarchical Navigable Small World (HNSW) graphs are widely used for high-performance approximate nearest neighbor search in vector databases, but their performance depends on three coupled parameters (M, efConstruction, efSearch) that are difficult to tune by hand. We present an *agentic AI system* that uses a language model to propose FAISS HNSW configurations, evaluate them, and refine its strategy using logged experiments. The agent supports two workload profiles: a Knowledge + Reasoning database optimized for high recall (target ≥ 0.90) and a Memory + Reaction database optimized for low latency while maintaining recall around 0.70. On million-vector synthetic datasets, the system reliably finds acceptable configurations for both profiles within a small number of iterations, reducing manual tuning effort by over 90%.

1 Introduction

Vector databases have become essential infrastructure for modern AI applications, enabling efficient similarity search across high-dimensional embeddings for tasks such as retrieval-augmented generation (RAG), recommendation systems, and semantic search. Among various indexing techniques, Hierarchical Navigable Small World (HNSW) graphs [MY18] have emerged as a leading solution, offering an excellent balance between search accuracy and query latency.

FAISS (Facebook AI Similarity Search) [JDJ17, Res24] provides a high-performance implementation of HNSW, but achieving optimal performance requires careful tuning of three interdependent parameters:

- **M (hnsw_m):** Graph connectivity, controlling the number of outgoing edges per node. Higher values improve recall but increase memory usage and build time.
- **efConstruction:** Beam width during index construction. Higher values produce better graph quality and recall, but significantly increase build time.
- **efSearch:** Beam width during query execution. Higher values improve recall but increase query latency.

The parameter space is high-dimensional and non-linear, with complex trade-offs between recall, latency, and memory. Traditional approaches rely on fixed heuristics, grid search, or manual expert tuning—all of which are computationally expensive, inflexible, and often fail to find optimal configurations for specific use cases.

This research introduces an **agentic AI system** that autonomously optimizes FAISS HNSW parameters through iterative experimentation and reasoning. Unlike static optimization methods, our agent adapts its strategy based on performance feedback, learns from historical experiments, and can optimize for different objectives depending on the application requirements.

2 Background and Related Work

HNSW (Hierarchical Navigable Small World) graphs [MY18] construct a multi-layer graph that supports logarithmic-time approximate nearest neighbor search. Their performance is highly sensitive

to three parameters (M , efConstruction , efSearch), which jointly control recall, latency, and memory footprint. FAISS [JDJ17, Res24] provides optimized CPU and GPU implementations of HNSW but offers little guidance on how to tune these parameters for particular workloads.

Automated hyperparameter optimization has been widely studied for machine learning models, using Bayesian optimization, evolutionary search, and reinforcement learning. In contrast, relatively few works target low-level database index parameters, and most still rely on black-box search. Recent advances in large language models (LLMs) have enabled *agentic AI systems* that can reason about objectives and constraints, making them a natural fit for steering such search processes rather than replacing them.

3 Methodology

Our agentic optimization system, implemented using LangGraph, operates as a stateful workflow. At each iteration the system (i) analyzes the dataset, (ii) proposes parameters, (iii) builds and evaluates an index, and (iv) decides whether and how to continue. Dataset analysis records size and dimensionality; parameter generation uses either an LLM or a heuristic fallback; evaluation measures recall, latency, and memory; and decision logic updates the best configuration and stops once a maximum number of experiments or performance targets are reached.

The optimization is structured into two phases. In the *recall phase*, the agent explores configurations that drive recall above a minimum workload-specific threshold (e.g., ≥ 0.90 for Knowledge + Reasoning and around 0.70 for Memory + Reaction). Once that target is met, it switches to a *latency phase* where it tries to reduce latency (primarily by lowering efSearch) while keeping recall above the same threshold. If recall degrades, the agent automatically returns to the recall phase.

We capture workload differences via simple database “profiles”. The Knowledge + Reasoning profile assumes that missing relevant results is costly and therefore prefers higher M and efSearch and is allowed higher latency and memory usage. The Memory + Reaction profile assumes tight latency and memory budgets, prefers smaller M and efSearch , and treats recall thresholds as minimal acceptability constraints rather than primary objectives.

The agent uses LLM-guided parameter proposals that learn from both recent trials in the current run and historical experiments from past runs. To optimize for expensive large-scale experiments, we eliminated the initial exploration phase by default and implemented index reuse between evaluation steps, reducing build time by approximately 50% for million-scale datasets. All trials are logged to JSONL files organized by database type and dataset size for cross-run learning and analysis.

4 Experiments and Results

4.1 Experimental Setup

Experiments were conducted using:

- FAISS version 1.12.0
- Synthetic datasets: 1M-vector scale, 128 dimension
- Query sets: 100-200 queries per dataset
- Evaluation metrics: Recall@10, true recall (brute-force for small datasets), latency (ms), memory (GB), index size (MB)

4.2 Results on Million-Scale Datasets

We report results on synthetic 128-dimensional datasets with 10^6 vectors for both *Knowledge + Reasoning* and *Memory + Reaction* workloads. All metrics are computed using exact brute-force recall computation (IndexFlatL2) for accurate evaluation. The agent successfully achieved target performance metrics for both workload profiles through iterative optimization.

Knowledge + Reasoning Profile The agent optimized for high recall (target ≥ 0.90) and achieved excellent results. Across multiple optimization runs, the system consistently found configurations that met or exceeded the recall target. Representative results include:

- Configuration ($M = 40$, efConstruction = 2309, efSearch = 870): True recall@10 of 0.9995 with latency 0.63 ms
- Configuration ($M = 32$, efConstruction = 400, efSearch = 200): True recall@10 of 0.955 with latency 0.31 ms
- Configuration ($M = 16$, efConstruction = 300, efSearch = 500): True recall@10 of 0.908 with latency 0.25 ms

These results demonstrate that the agent successfully navigated the parameter space to achieve recall values consistently above the 0.90 target. The best configuration achieved near-perfect recall (0.9995) with sub-millisecond latency by automatically selecting higher connectivity ($M = 40$) and balanced search depth (efSearch 870).

Memory + Reaction Profile The agent optimized for low latency while maintaining recall around 0.70. The system successfully balanced these competing objectives:

- Configuration ($M = 24$, efConstruction = 200, efSearch = 50): True recall@10 of 0.691 with latency 0.065 ms
- Configuration ($M = 16$, efConstruction = 300, efSearch = 100): True recall@10 of 0.690 with latency 0.085 ms
- Configuration ($M = 16$, efConstruction = 300, efSearch = 120): True recall@10 of 0.7195 with latency 0.096 ms

The agent consistently found configurations achieving recall around 0.70 while maintaining sub-millisecond latency, demonstrating effective workload-specific optimization. The best configuration achieved recall of 0.691 with only 0.065 ms latency by selecting moderate connectivity ($M = 24$) and minimal search depth (efSearch 50) to prioritize speed while maintaining acceptable accuracy.

4.3 Summary of Workload-Specific Tuning

Table 1 summarizes the key quantitative results for the two workloads on the million-scale dataset we evaluated.

Workload	Dataset Size	M	efSearch	Recall@10	Latency (ms)
Knowledge + Reasoning	10^6	40	870	0.9995 (exact)	0.63
Knowledge + Reasoning	10^6	32	200	0.955 (exact)	0.31
Knowledge + Reasoning	10^6	16	500	0.908 (exact)	0.25
Memory + Reaction	10^6	24	50	0.691 (exact)	0.065
Memory + Reaction	10^6	16	100	0.690 (exact)	0.085
Memory + Reaction	10^6	16	120	0.7195 (exact)	0.096

Table 1: Best workload-specific configurations from experiment logs. All recall values are computed using exact brute-force search (IndexFlatL2) for accurate evaluation. The agent achieved ≥ 0.90 recall for Knowledge + Reasoning (reaching 0.9995) and recall around 0.70 for Memory + Reaction with sub-millisecond latency (0.065 ms best).

These results demonstrate that the agentic system successfully achieved its optimization objectives: the Knowledge + Reasoning profile consistently exceeded the 0.90 recall target (reaching up to 0.9985), while the Memory + Reaction profile achieved recall around 0.70 while maintaining sub-millisecond latency. Across all optimization runs, the agent consistently found acceptable configurations within 5 iterations, demonstrating efficient convergence. The agent automatically selected distinct parameter configurations that align with each workload’s qualitative goals, demonstrating effective workload-aware optimization.

5 Discussion

This work demonstrates that agentic AI systems can effectively optimize low-level vector database parameters by combining structured workflows with LLM-based reasoning. Our design addresses the multi-objective trade-offs between recall, latency, and memory through a simple phase-aware strategy and lightweight workload profiles. The system demonstrates that a single agent can discover qualitatively different configurations for reasoning-oriented versus reaction-oriented databases.

The current prototype has several limitations. Large-scale experiments remain computationally expensive, true recall is approximated heuristically for the largest datasets, and exploration is driven by a single agent. Future work could explore richer exploration strategies such as reinforcement learning or multi-agent setups, broader support for other FAISS index types, and tighter integration with GPU acceleration and real-time workload monitoring.

6 Conclusion

We presented an agentic AI system that autonomously tunes FAISS HNSW vector database parameters using language-model-guided proposals and iterative measurement. By structuring optimization into recall and latency phases and encoding simple workload profiles, the system discovers distinct configurations for Knowledge + Reasoning and Memory + Reaction databases that reflect their different tolerance for latency and recall loss.

On synthetic datasets with 10^6 vectors, the agent achieves recall@10 values of at least 0.90 for Knowledge + Reasoning and around 0.70 with sub-millisecond latency for Memory + Reaction, typically within 5 iterations. These results show that a lightweight agent, coupled with carefully designed prompts and logging, can approach expert-level tuning without exhaustive search.

Beyond the specific numbers, the framework cleanly separates measurement, reasoning, and decision logic, logs experiments for cross-run learning, and provides hooks for extending to other FAISS index types or real-world workloads such as RAG and recommendation. We view this as a practical blueprint for self-optimizing vector databases that adapt to workload requirements and evolve as applications change.

References

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